

Supplement to “Benign Exclusion and Consensus-Test Limits in Heterogeneous Federated Learning”

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TABLE XV
BENIGN UTILITY COST OVER PAIRED DATASET-SEED UNITS
(PERCENTAGE POINTS)

α	Rule	Pairs	Mean	Min.	Max.
0.5	Norm clipping	9	+0.08	−0.07	+0.78
0.5	Coordinate-wise median	9	+3.76	+0.57	+19.07
0.5	Multi-Krum	9	+1.70	−2.71	+10.28
0.5	FLTrust (normalized)	9	+6.02	+1.73	+16.56
0.5	FLAME (HDBSCAN)	9	+7.76	+1.44	+36.85
0.5	Fed-MDBSCAN-G (full)	9	+0.60	−1.22	+6.52
0.1	Norm clipping	9	+0.71	−0.54	+5.40
0.1	Coordinate-wise median	9	+14.37	+4.56	+43.57
0.1	Multi-Krum	9	+12.89	+4.05	+48.49
0.1	FLTrust (normalized)	9	+12.19	+2.77	+33.29
0.1	FLAME (HDBSCAN)	12	+17.66	+6.12	+48.52
0.1	Fed-MDBSCAN-G (full)	12	+6.73	−0.06	+48.42
0.01	Norm clipping	9	+11.92	−2.34	+36.62
0.01	Coordinate-wise median	9	+43.03	+19.61	+67.85
0.01	Multi-Krum	9	+29.83	+2.31	+51.50
0.01	FLTrust (normalized)	9	+9.08	−1.31	+19.19
0.01	FLAME (HDBSCAN)	12	+35.55	+6.92	+68.47
0.01	Fed-MDBSCAN-G (full)	12	+31.60	−0.14	+62.40

Minima and maxima span datasets and seeds, not confidence intervals. The artifact includes each per-seed difference. Pairing checks dataset, partition, initial model and participation schedule hashes. Coverage varies by rule and concentration.

I. PAIRED BENIGN UTILITY MEASUREMENTS

Table XV summarizes the individual differences used in the adversary-free utility analysis. A pair matches the dataset, concentration, seed, data partition, initial model, and participation schedule. Differences are uniform-mean accuracy minus rule accuracy. The range spans the available dataset-seed pairs; it is neither a confidence interval nor a claim about statistical significance. The artifact file `benign_paired_seed_deltas.csv` retains all individual differences.

II. DUPLICATE-VECTOR DENSITY CHECK

The deployed density function returns 1 if the sum of effective nearest-neighbour distances is exactly zero. This is distinct from the ideal reciprocal-distance expression at strictly positive sums. With $[0, 0, 0.1, 0.2]$ and $k = 1$, it returns $[1, 1, 10, 10]$. Six coincident observations satisfy the zero-sum condition at $k = 5$; a duplicated pair with four distinct neighbours need not. Four small inputs were evaluated in the working source and two frozen sources, giving 12 passing checks. The function ASTs agree. These checks verify the lo-

cal boundary rule, not equivalence to the original MDBSCAN authors’ implementation.

III. FLAME IDENTITY COUNTS AND SUBSEQUENT INSTRUMENTATION

All 201 canonical FLAME result files match the hashes recorded in the earlier local audit: 200 completed runs and one retained CIFAR failure. The 6,000 completed rounds each contain 90 participants and between 46 and 89 accepted identities. Participant, accepted and rejected identities form disjoint complete partitions, and their confusion counts agree with the exported cell totals. No recorded round uses fallback or accepts everyone. In the frozen selector, the small-cohort and zero-distance skip paths accept everyone, so those paths are excluded by the recorded decisions. Normal selection keeps indices with nonnegative HDBSCAN labels; its minimum cluster size of 46 permits at most one cluster among 90 participants. This establishes the association between saved acceptance-set size and the retained cluster size for these records. It does not reconstruct missing historical labels.

Subsequent instrumentation records whether clustering executed, its skip reason, the configured minimum cluster size, label-derived cluster sizes, the noise count, and the selection size before any fallback. A skipped fit records unknown sizes; an executed all-noise fit records an empty cluster dictionary. The full simulation test suite passes 125 tests. Six bounded frozen/current selector comparisons through the same server preserve all previous non-timing decision fields and produce identical model weights. These checks do not add research training runs or alter the archived results.

IV. CHECKPOINT IDENTITY AND SCOPE

The radius check uses 72 distinct recorded matrices and two gate evaluations per matrix. Alternative gate and cutoff settings were evaluated offline on the saved matrices and were never fed back into training. Its 36 attacked matrices contain 18 adversarial clients each. All admit the full cohort at the initial stage. The sufficient consensus condition holds in 35, with maximum ratio 2.017225 at the one exception. Gate rows are deduplicated by reference and round before aggregation. With the neighbourhood cutoff enabled, six of the 144 gate evaluations contain candidate groups with members outside the initial pool. All occur outside the attacked block. The subset hypothesis therefore requires an explicit check. Three additional historical batch-regime rows in the supplied radius

CSV are not counted as members of the 36-matrix attacked sample.

V. IMPLEMENTATION CHECKS

FLTrust. The local aggregation rule was compared with a transcription of the rule in the authors’ demonstration code on 18 recorded matrices and four root constructions; the relative discrepancy between aggregated updates was below 2.8×10^{-7} . Degenerate-root and zero-trust paths diverge. Root norms were not recorded in the 5,490 canonical rounds, so an unflagged degenerate-root path cannot be excluded in the 135 rounds that accepted every submission. Root and client local-iteration counts also differ, so the check does not establish fidelity of the full training protocol.

FLAME. Six small checks exercise clipping, averaging, and noise scaling through the server’s aggregation path. Equivalence of the HDBSCAN clustering with the authors’ implementation, and the original noise calibration, remain unverified.

Multi-Krum. The local selection was compared with the rule of Blanchard *et al.* on six Fashion-MNIST Min-Max checkpoint matrices of the decision-path replay (seeds 137 and 2024, rounds 0, 9, and 29), through the server’s aggregation path. The server derives $f = 27$, 61 neighbours, and $m = 61$, and its distance computation agrees with an independent float64 computation of the published score to a relative difference of 1.65×10^{-13} . In four of the six matrices the two selected sets have a symmetric difference of four to six members. All differing members belong to a block of 18 byte-identical attacker updates whose tied scores straddle the selection cutoff; the published rule does not specify tie-breaking. The aggregated updates agree to a relative difference below 2.6×10^{-8} in all six matrices, including the two with identical selections. Across the 5,130 canonical Multi-Krum rounds, the recorded cohort is 90, the validity condition $2f + 2 < n$ holds, 61 updates are selected in every round, and no fallback is flagged. The canonical update matrices were not saved, so their scores were not recomputed. The authors’ code was not run, and no multi-round trajectory was compared. The published experiments use $m = n - f$; with $m = 63$ the selected sets would share 59 to 61 of the 61 members.

Cardinality identities. A rule that retained the wrong 61 updates would still have $\text{BER} = 29/90$ in a clean round, so agreement with a cardinality identity is no evidence that its scores or selected identities reproduce the original implementation.

Multi-Krum under the constrained probe. In five of the six cells the confusion counts are $\text{TP} = 0$, $\text{FP} = 2,610$, $\text{TN} = 3,060$, and $\text{FN} = 2,430$.

VI. CHECKPOINT DETAILS MOVED FROM THE MAIN TEXT

The median radial ratio over the 36 attacked matrices is 1.448095 and the maximum is 2.017225, at Fashion-MNIST, seed 137, patch, round 29. Promotion to float64 changes Γ by at most 5.43×10^{-7} relative to the native computation and does not alter any threshold classification. Evaluating two gates yields 72 rows from the same 36 matrices, which does

not double the sample size. Because the initial and consensus tests use different centres and radii, the convex-hull bound alone does not order their client-level decisions. The 72 replay matrices cover three seeds and rounds 0, 9, and 29. At the clean MNIST checkpoint of the batch-regime example, final rejections are 34, 25, and 18 in regimes A, B, and C; of the 16 clients rejected in A but accepted in C, 10 no longer belong to a tested group and six change at Stage 1. Policy P1 rejects 44 clients in all three regimes, although one pair of identities differs in regime C.

VII. EXTENDED TABLE NOTES

The notes under the main-text tables give only the definitions needed to read each table. The remaining remarks from earlier versions of those notes are collected here; no result has been removed.

Table III (benign exclusion). Multi-Krum’s BER is $1 - m/n = 29/90 = 32.22\%$ in every cell, independent of dataset and of α (Corollary 1). FLAME’s clean upper bound is $1 - (\lfloor n/2 \rfloor + 1)/n = 44/90 = 48.89\%$ (Corollary 2). CIFAR-10 has no $\alpha = 0.5$ block. FLTrust’s zero-weight count is not a separate rejection stage.

Table IV (benign utility cost). Entries that round to zero are printed without sign. Per-seed paired differences are supplied in the artifact, and Table XV summarizes their ranges. No significance claim is attached to any single entry.

Table V (FLAME cluster sizes). The frozen normal path retains one non-noise cluster; historical cluster labels were not saved. The minimum size is attained in 719/990 rounds. The counts are not reconstructed from BER.

Table VI (round-level alarm). At $\alpha = 0.01$ every adversary-free round on every dataset raises an alarm. The counts measure false alarms in clean controls, not predictive value at a deployment prevalence. No zero rate is reported for the other rules, because absence of an alarm mechanism is not alarm specificity.

Tables VII and VIII (discrimination). $J > 0$ lies above the chance diagonal. The pooled cells are not independent replicates, and family coverage differs in dataset, concentration and adversary ratio. The summary does not measure radial displacement and is not a controlled displacement experiment. FLTrust has a positive mean J for the constrained family; its counts concern zero weights, and this does not establish a general distinction between defense classes.

Table IX (constrained adversary). Only the two rules whose admission arithmetic yields a numerical prediction are listed. Multi-Krum discards $n - m = 29$ of 90 submissions every round; in 5 of the six cells it records $\text{TPR} = 0$ with exactly $2,610 = 29 \times 90$ honest rejections, i.e. $\text{FPR} = 29/63 = 46.03\%$, so every discarded submission is honest. The CIFAR-10/FLAME/ $\alpha = 0.1$ cell also has one retained failure. The conditional identity does not apply to the positive-TPR rows.

Table XII (coverage replay). Widening coverage adds honest rejections and, except in one block, no attacker rejections. That block (P2, density-only gate, patch) reaches $\text{TPR} = 46/324 = 14.20\%$ at $\text{FPR} = 142/1296 = 10.96\%$, i.e. $J = +3.2$ pp; its attack-disabled control records 82 honest rejections under the

same policy. Under the density-only gate, P2 would exclude all 1,296 honest Min-Max observations, and the valve restores B_0 in all 18 of those matrices. These are fixed-matrix decision counts; no accuracy or attack success rate is re-measured.

Table XIII (recovered clusters). The decision column is the implementation’s own output. The majority cluster lies at 0.12–0.13 of the rejection radius and every honest minority cluster at 1.70–2.63 times it, so the rejection radius separates honest groups from one another.

Table XIV (ablation). Removing every layer above the stage-1 acceptance region changes the mean by less than a tenth of a percentage point; no equivalence claim is made from a small mean difference.

VIII. REPRODUCTION BOUNDARY

From the V5 submission directory, run `python analysis/build_tables.py --root .` with the existing project Python environment. It checks the copied report hashes and condition metadata, regenerates the tables and figures, and verifies the direct-count and checkpoint summaries. The main and supplement are compiled with the supplied IEEEtran files. A repository clone suffices for these summary-level checks. Repeating the historical identity-level audit or recapturing training checkpoints requires the separate local raw archive and frozen environment. Hash equality establishes content integrity, not a hypothesis’s timing or preregistration.

The five failed canonical units remain failures. Deterministic CIFAR diagnostics are not replacements for the failed unit. No new backdoor improvement, statistical equivalence, or general defense ranking is inferred from the analyses in this supplement.